STAT 143, Introduction to Probability and Statistics

Syllabus, Fall 2026

Course Information.

- 4 semester hours
- Instructor: Thomas Scofield
- Texts:
  - *Statistics Using Technology*, 1st Edition, by Kathryn Kozak, perhaps sometimes referred to as **SUT1**. We will cover all chapters, with perhaps a few sporadic items added, and a few others left out.
  - *Social Justice Fallacies*, by Thomas Sowell
- Class meetings: MWF, 8:00–9:05 am, NH 295

Catalog Description.
An introduction to the concepts and methods of probability and statistics. The course is designed for students interested in the application of probability and statistics in business, economics, and the social and life sciences. Topics include descriptive statistics, probability theory, random variables and probability distributions, sampling distributions, point and interval estimation, hypothesis testing, analysis of variance, and correlation and regression. No student may receive credit for both STAT 143 and STAT 145.

The course meets the Mathematical Sciences Core requirement.

Student Learning Outcomes. Upon completion of this course, students will be able to

- Explain basic principles of study design, describing their role in answering research questions.
- Produce appropriate graphical and numerical summaries of one or two variables (categorical and/or quantitative).
- Use confidence intervals and hypothesis tests to make inferences about a population based on a sample drawn from the population.
- Choose an appropriate statistical model to analyze data in certain common situations.
- Verify whether underlying assumptions justifying the use of a statistical model are met.
- Critically evaluate presentations of statistical results (for example, in journal articles, media pieces, case studies, etc.)
- Use statistical software in the pursuit of the outcomes listed above.

As this is a core course, we formally expect students to

- Apply algorithmic, statistical, and/or mathematical methods to solve problems, broadly defined to find the answers to questions in various domains (as appropriate).

- Articulate limiting assumptions or limitations to the conclusions that can be drawn from the use of these methods and identify appropriate and inappropriate uses of such methods, as informed by a Reformed Christian perspective.
- Employ data-driven, mathematical, statistical, and/or software models, analyzing their results to answer questions, solve problems, support arguments, draw conclusions, make predictions, and/or identify possible causal relationships.
- Identify and use appropriate mathematical and statistical tools for solving a given problem; implementing solutions using StatCrunch software, but with an ability to explain the algorithms used.
- Represent, interpret, and process information in graphical, numeric, and/or symbolic forms.

Student achievement with regard to these outcomes will be assessed via reflections, quizzes and test questions.

Topics include

1. Structure and organization of data
2. Sampling and study/experimental design
3. Graphical and numerical summaries of data
4. Basic probability theory
5. Probability distributions, and methods for simulating them; Central Limit Theorem
6. Parameter estimation/confidence intervals, taken from settings such as 1-proportion, single mean, difference of means, correlation/slope
7. Null-hypothesis significance testing, taken from settings such as univariate (goodness-of-fit, 1-sample t) and bivariate data (2-sample t, chi-square, 1-way ANOVA, model utility)
8. Simple linear regression
9. Multiple regression (if time allows)

Methods of Evaluation.

Assessment	Pct
Reflections on readings/prompts	10%
Quizzes	14%
Midterms (Sept. 30, Nov. 4, and Dec. 4)	51%
Final (Dec. 14, at 1:30 pm)	25%

Policies.

- You are expected to attend class faithfully, in person. When you cannot, regardless of reason, you are responsible for catching yourself up.
- Before class begins, visit the restroom, and prepare yourself for a prompt beginning. Except on specified days, this includes parking your electronic devices.
- Written work, when assigned, will be in the form of reflections on assigned readings, providing your understanding of a concept, and/or giving examples of a concept. The

purpose is to verbalize your own response and processing of course ideas, not to have you echo words of a classmate, nor to get an AI-influenced response. Though it is helpful to discuss ideas with classmates, you must submit work composed by you alone. It should contain not only final answers, but a Violations of this policy shall result in a score of zero. Repeated instances shall result in a course grade of ”F”.

- Course administration (communication, assignments, scores, etc.) is conducted in MyOpenMath (MOM). Check the MOM calendar regularly for daily readings and learning objectives, assignments, and anything else deemed useful. Stay on top of assignments and their due dates. If you should miss a deadline, you have a bank of 6 late passes that, when used, extend a deadline (usually) 48 hours. **You may not use more than one late pass per assignment.** Be a good steward of your late passes as, almost certainly, there will be times when circumstances beyond your control are the cause of lateness.
- You are expected to take exams on the dates specified, or provide sufficient cause why you cannot. Family trips, pre-arranged flights, etc. are *not* sufficient.
- Basic calculators with only a few functions are supplied by the instructor, serving as the lone computing device allowable during evaluations.

Artificial Intelligence. When taking quizzes and exams, no AI is allowed. When reflecting on a reading or prompt, AI is not to be used. In both these instances, questions are to be answered in a manner that reveals what **you** know, what **you** have learned on a subject, **your** understanding of concepts and nomenclature in the field of statistics, and **your** ability to apply those in real-world settings. Do not use AI for any of the items in this course which are used directly to determine your grade.

To wrestle with problems, to err and then learn from your mistakes, to synthesize ideas in a coherent way so you can verbalize them to another human, to become personally adept at mimicking your instructor's choices of statistical tools tailored to different settings so as to draw reasonable conclusions from data—these are formative, human experiences, and hallmarks of the value of a university education. They bring the joy of discovery, the satisfaction of having agency to make sense of God's miraculous, oft-times-surprising world. There seem to be more ways AI can rob you of this value than ways for it to usefully enhance it. If you use AI at all, it may be best to limit that use to the generation of practice questions, beyond those I give, as helps for understanding a difficult concept, or as drills prior to an examination.

Accommodations. Calvin University is committed to providing reasonable accommodations for students with disabilities. Students with a documented disability should notify a Disability Coordinator in the Student Success Office (HH 227) to discuss appropriate accommodations. If you have an accommodation memo, talk with me early about arrangements, preferably within the first 2 weeks of the semester.

Exceptions. I reserve the right to make changes or exceptions to course policies, including those described in this document, either for the entire class or for individuals. The ultimate goal in this course is **learning**, and formal requirements should not unnecessarily stand in the way of that. Thus, if you think that any of the conditions of the course are interfering with learning, please speak with me about this, and we will consider what can be done.